SCIENCE & INNOVATION FOR SUSTAINABILITY

FOUR PRESENTATION POSTER BLUEPRINTS & SPEAKING NOTES

Document 05: Step-by-Step Handwriting Guides & Oral Presentation Scripts for 4 Exhibition Posters

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Important Instructions for Student Presenters: *For science exhibitions, fairs, and innovation competitions, presenters typically prepare* ***four handmade display charts*** *on standard chart paper (55 cm x 70 cm). Do NOT paste printed A4 sheets on your charts! Judges love hand-drawn posters made with sketch pens, markers, and rulers. This guide gives you the exact text to write, the exact drawings to sketch, and the exact protractor angles for your pie chart. We also wrote down our practice speaking script for each poster so you can explain your work clearly in 2 minutes.*

POSTER 1: THE PROBLEM AND THE IDEA

Chart Sheet Size:	Standard Chart Paper (55 cm x 70 cm), White background
Pen Colors to Use:	Main Heading: Bold Navy Blue | Subheadings: Forest Green | Text: Black marker
Sheet Space Split:	Top Banner: 15% | Left Side (The Story & Problem): 42% | Right Side (Aim & Plant Disease Triangle): 43%

Exact Words and Layout to Write on Poster 1:

1. TOP BANNER (Big Bold Letters across the top):

KRISHI-DRISHTI: Offline-First Edge-AI Plant Health Screening Wand

Theme: Science and Innovation for Sustainability | Student Research Project

2. TRACKING REAL-WORLD CROP LOSSES (Left Side, Upper Half):

• Last year, viral agricultural news and social media showed devastating crop failures across major tomato and potato belts.
• In tomato hubs like Kolar and Nashik, early blight destroyed harvests, forcing farmers to dump damaged crates on highways.
• Weeks later, winter fog hit northern India, and potato fields in Samastipur, Vaishali, and Agra turned black from Late Blight in under 72 hours!
• Seeing the exact same tragedy repeat across Place A, Place B, Place C, and Place D, we investigated why farmers couldn't catch blights early.

3. WHY PRESENT SOLUTIONS DON'T WORK FOR SMALL FARMERS (Left Side, Lower Half):

• *No 4G Signal in Muddy Fields:* Commercial smartphone apps need cloud internet and hang indefinitely in rural fields.
• *Costly Diagnostic Labs:* Testing at a university lab costs thousands of rupees and takes 10 to 14 days (too late!).
• *Panic Chemical Spraying:* Farmers spray expensive chemical cocktails blindly, wasting money and ruining soil.

4. OUR RESEARCH QUESTION & AIM (Right Side, Upper Half):

• **Our Question:** Can a student-built handheld wand under Rs 1,500 that uses an ESP32, an 8 cm optical hood, and an offline phone provide instant leaf health screening without internet?
• **Our Aim:** Build a cheap, pocket-sized screening kit to help marginal farmers detect blights early and spray safely.

5. CORE BIOLOGY: THE PLANT DISEASE TRIANGLE (Right Side, Lower Half):

[Draw a large green triangle with your ruler and sketch pens]:

Top Vertex: Susceptible Host (Potato or Tomato Plant)
Left Vertex: Virulent Pathogen (Water-mold spores in air)
Right Vertex: Conducive Environment (Relative Humidity > 85%, Temp 18–24.5°C)

• *Why this matters:* Without 85% humidity, spores cannot germinate! Our wand checks canopy humidity so farmers don't spray during dry weather.

What to Say to the Judges for Poster 1 (2-Minute Talk):

"Respected judges and reviewers! Last year, while following agricultural news and farmer channels on social media, I saw shocking videos of tomato and potato crops getting destroyed across multiple states. First, farmers in Kolar and Nashik were dumping crates of rotten tomatoes because early blight ruined their harvests. Then, during winter fog, potato fields across Samastipur, Vaishali, and Agra turned black from Late Blight in just three days! In farmer comments, hundreds asked what to spray, but by then spraying was useless. Existing apps failed because 4G signals vanish in open fields, and lab tests cost thousands and take two weeks. So we asked: why can't we build an affordable testing wand under Rs 1,500 that works completely offline right in the dirt? That is how KRISHI-DRISHTI was born."

POSTER 2: HARDWARE DESIGN & HOW IT WORKS

Chart Sheet Size:	Standard Chart Paper (55 cm x 70 cm), White background
Pen Colors to Use:	Main Heading: Navy Blue | Circuit Lines: Dark Blue & Black | Callouts: Forest Green
Sheet Space Split:	Top Banner: 12% | Left Side (Hardware Diagram & Parts Table): 48% | Right Side (4-Page Screen & Data Flow): 52%

Exact Words and Layout to Write on Poster 2:

1. MAIN HEADER (Large Bold Letters, Top Center):
SYSTEM HARDWARE & PHYSICAL WORKING PRINCIPLE

2. THREE PHYSICAL MODULES (Left Side, Top):
• *Unit 1: Handheld Controller Wand:* ESP32 DevKit V1 + OLED Display + DHT11 humidity sensor + SCAN/NEXT buttons.
• *Unit 2: Optical Camera Wand:* AI-Thinker ESP32-CAM + 5MP OV5640 sensor + 8 cm black optical hood + diffuse LED.
• *Unit 3: Pocket Edge-AI Engine:* Old Android phone running offline Python daemon over local Wi-Fi hotspot (192.168.4.1).

3. HARDWARE BILL OF MATERIALS TABLE (Left Side, Bottom):
[Draw this neat table with a ruler]:
• ESP32 DevKit V1: Rs 420 | AI-Thinker ESP32-CAM: Rs 540 | SSD1306 OLED: Rs 190
• DHT11 Sensor: Rs 95 | 16GB MicroSD: Rs 160 | 8 cm Hood: Rs 35 | Wires/Capacitor: Rs 30
• **Total Hardware Cost = Rs 1,480** (Affordable to any small farmer cooperative)

4. STEP-BY-STEP DATA FLOW (Right Side, Top):
[Farmer Clicks SCAN] --> [Wand Reads DHT11] --> [Wi-Fi GET /api/capture]
--> [Camera Snaps UXGA Photo inside Shroud] --> [Phone Fetches Photo]
--> [MobileNetV3 TFLite Runs (4.9 ms)] --> [Checks Shannon Entropy]
--> [Uploads Diagnosis] --> [Wand Displays 4 Screens on OLED]

5. THE 4-PAGE OLED SCREEN CAROUSEL (Right Side, Bottom):
[Draw 4 small rectangles representing the 128x64 OLED display]:
• **Screen 1 (Diagnosis):** POTATO: LATE BLIGHT [89%] (or UNCERTAIN / RESCAN)
• **Screen 2 (Microclimate):** TEMP: 21.4°C | RH: 88% | HIGH BLIGHT RISK
• **Screen 3 (Prescription):** SPRAY TRICHODERMA OR COPPER OXYCHLORIDE (2g/L)
• **Screen 4 (Telemetry):** BATT: 4.12V | RSSI: -48dBm | SD: 42 SCANS

What to Say to the Judges for Poster 2 (2-Minute Talk):

"Respected judges, let me show you the actual physical hardware. It consists of two wands and an ordinary phone. In my right hand is the controller wand with an OLED screen and a humidity sensor. In my left hand is the camera wand. Notice this black 8 cm hood on the camera lens. Without this tube, if a farmer holds a phone, the photo is blurry, and blinding sun glare tricks the AI. When we place this tube against the leaf, it locks the focus distance at exactly 8 cm and completely blocks harsh sunlight. The camera creates its own local Wi-Fi hotspot. My old phone in my pocket grabs the picture, runs our neural network in just 4.9 milliseconds, and sends the answer back to the OLED screen. Zero internet, zero cloud servers, and total cost is only Rs 1,480!"

POSTER 3: EXPERIMENTS, BENCHMARKS & PIE CHART

Chart Sheet Size:	Standard Chart Paper (55 cm x 70 cm), White background
Pen Colors to Use:	Main Heading: Navy Blue | Bar Chart: Blue & Green | Pie Chart: Green, Yellow & Red
Sheet Space Split:	Top Banner: 12% | Left Side (Dataset Audit & Clean Results): 48% | Right Side (Pie Chart with Exact Angles): 52%

Exact Words and Layout to Write on Poster 3:

1. MAIN HEADER (Large Bold Letters, Top Center):
EXPERIMENTAL VALIDATION & BENCHMARK PERFORMANCE

2. ZERO-LEAKAGE DATASET AUDIT (Left Side, Top):
• We audited 7,710 records and reconciled them into exactly **6,965 unique cryptographic hashes**.
• We discovered 5 duplicate image pairs in the public Kaggle dataset—all verified to share the same label (0 label conflicts).
• Guaranteed zero data leakage: our test benchmark leaves were 100% unseen by the neural network during training.

3. OUR TWO MAIN BENCHMARK RESULTS (Left Side, Bottom):
• *Clean Botanical Benchmark (60 Leaves):* **88.33% strict accuracy (53/60)** | **92.98% accepted accuracy (53/57)**.
• *Pathogen Family Match:* 54 out of 60 correct (**90.0%**) — recognizing the underlying oomycete water-mold.
• *Tricky Outdoor PlantDoc Images (93 Photos):* Raw baseline accuracy was poor (**33.33%, 31/93**). But when our Shannon entropy filter was turned on, it safely caught 37 confusing photos, raising accepted accuracy to **55.36% (31/56)**!

4. EXACT PROTRACTOR DRAWING GUIDE FOR OUR PIE CHART (Right Side):
[Draw a large circle with a compass (Radius = 7 cm). Use a school protractor to measure these exact angles]:
• **Sector 1 (Accepted Correct Diagnosis):** 31 / 93 = 33.33% --> **Exact Protractor Angle = 120°** [Color: Green]
• **Sector 2 (Gated Uncertain / Safe Rescan):** 37 / 93 = 39.78% --> **Exact Protractor Angle = 143°** [Color: Amber Yellow]
• **Sector 3 (Accepted Misclassification):** 25 / 93 = 26.88% --> **Exact Protractor Angle = 97°** [Color: Light Red]
• **Math Verification Check:** 120° + 143° + 97° = **Exactly 360°** (A complete perfect circle!)
Student Drawing Tip: Start with a flat horizontal line from center. Measure 120° for Sector 1. From that second line, measure 143° for Sector 2. The remaining slice will automatically be exactly 97°!

What to Say to the Judges for Poster 3 (2-Minute Talk):

"Respected judges, many AI projects claim 99% accuracy because they test on images the computer already memorized. We wanted our science to be 100% honest. We used cryptographic hashes to guarantee that our 60 botanical benchmark leaves were completely unseen. On those clean leaves, we scored 88.33% strict diagnosis and 90.0% pathogen match. Then, to stress-test our system, we tested 93 messy, outdoor field photos from the PlantDoc collection. Raw accuracy was only 33.33% because real field dirt confuses AI. But look at this pie chart: our Shannon entropy filter safely caught 37 confusing images and marked them as 'UNCERTAIN' instead of guessing blindly. That raised our accepted accuracy to 55.36%. In farming, an honest 'I am unsure, please rescan' is ten times better than a wrong chemical recommendation!"

POSTER 4: LIMITATIONS, FARMER IMPACT & FUTURE GOALS

Chart Sheet Size:	Standard Chart Paper (55 cm x 70 cm), White background
Pen Colors to Use:	Main Heading: Navy Blue | Limitations: Warm Amber | Impact & Future: Forest Green
Sheet Space Split:	Top Banner: 12% | Left Side (Honest Failure Cases & What Broke): 48% | Right Side (Sustainability & What's Next): 52%

Exact Words and Layout to Write on Poster 4:

1. MAIN HEADER (Large Bold Letters, Top Center):
REAL-WORLD LIMITATIONS, SUSTAINABILITY & FUTURE ROADMAP

2. WHAT WORKED REALLY WELL (Left Side, Top):
• *100% Offline Mesh:* Complete diagnosis without internet, mobile towers, or SIM cards.
• *4.9 ms Latency:* Near-instant neural inference on an ordinary budget phone in the pocket.
• *Long Battery Life:* Ran for 6 hours 42 minutes (> 400 scans) on a simple 5,000 mAh power bank.
• *Microclimate Spray Suppression:* Stopped unnecessary spraying when humidity was low (< 85%).

3. HONEST MISTAKES AND REAL LIMITATIONS (Left Side, Bottom):
• *The Alien Tea Leaf Trap:* When we put an unseen tea leaf inside the hood, the model called it 'Potato Late Blight' with 87% confidence! This proved our shroud standardizes distance, but cannot turn a simple AI into a botanist. Farmers must only scan potato or tomato leaves.
• *Drop-Shadow Glitch:* An angled flashlight created dark shadows behind leaf veins that tricked the AI into seeing blight. We fixed this with a diffuser disc.

4. SUSTAINABILITY & IMPACT ON MARGINAL FARMERS (Right Side, Top):
• *Big Cost Savings:* Preventing just one wasted fungicide spray saves Rs 1,500 to Rs 2,500 per acre—paying for the entire wand in one week!
• *Soil & Bee Health:* Promotes organic Trichoderma treatments; stops toxic chemical runoff into village groundwater.
• *Community Sharing:* One wand can be shared by an entire self-help group (SHG) or village cooperative.

5. WHAT WE PLAN TO BUILD NEXT (Right Side, Bottom):
• *On-Chip AI:* Run an int8 neural model directly on the ESP32 chip (using ESP-NN), so no smartphone is needed at all!
• *More Winter Crops:* Train the model to recognize wheat rust and mustard blight.
• *Audio Advice in Local Dialects:* Add a small speaker that speaks advice in local languages so farmers can listen easily.

Closing Presentation Speaking Script for Poster 4 (Final 2-Minute Pitch):

"To wrap up our presentation, respected judges: KRISHI-DRISHTI is not a fantasy project made from internet cut-outs. It is a working, physical engineering tool built with our own hands in our school lab. We showed you our exact circuits, our code, our battery life, and our honest failures. We don't claim our wand is a magical device—we openly showed how a tea leaf tricked it and how night shadows caused false alarms. But by combining low-cost microcontrollers, an 8 cm optical hood, and canopy humidity sensing, we proved that smallholder farmers can screen their crops for under Rs 1,500 without needing internet. This saves them money, protects our soil, and puts practical science right into the farmer's hands. Thank you so much!"